1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
import numpy as np
import pandas as pd
import subprocess
import matplotlib.pyplot as plt
import os
import sys

from tqdm import tqdm
from pathlib import Path
from torch.utils.data import Dataset, DataLoader


# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkp
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, ge

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)
```

```
# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
==================================================
 CONFIGURACIÓN INICIAL
==================================================
Configuración Valor
       Device  cuda
         Seed    42
 Output Width   100
==================================================
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'p-annealing',        # P-annealing (Lp^p con p decreciente)
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 100000,                # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',  # Path base para cl
    'experiment_base_path': os.path.abspath('../../../experiments') ,      # sae/experiments/
    'extras': {
        'expansion': 16,
        'sparsity_coef': 5e-1,
        'epochs': 200,
        'early_stopping': False,
```

```python
        'learning_rate': 1e-3,
        'p_start': 1.0,
        'p_end': 0.2,
        'n_sparsity_updates': 10,
        'anneal_start': 40000,
        'batch_size': 2048,
        'warmup_steps': 8000,
        'sparsity_warmup_steps': 16000,
        'estimated_epochs': 200
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata de naming:
  variante: standard
  tecnica: p-annealing
  capa: 6
  juegos: 100000
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 16, 'sparsity_coef': 0.5, 'epochs': 200, 'early_stopping': False, 'le
```

```python
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')
```

```
extras_id: ex4
```

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```
# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 8192,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 2048,
    'num_epochs': 200,
    'learning_rate': 1e-3,
    'warmup_steps': 8000,        # steps de LR warmup lineal
    'sparsity_warmup_steps': 16000, # steps de rampa de sparsity (0→1)
    'sparsity_coef': 5e-1,       # alpha inicial (loss sum(dim=-1).mean())

    # P-annealing
    'p_start': 1.0,              # p inicial (equivalente a L1)
    'p_end': 0.2,                # p final
    'n_sparsity_updates': 10,    # número de saltos discretos
    'anneal_start': 40000,         # step en que empieza el annealing (después del warmup)
    'estimated_epochs': 200,       # épocas estimadas para condensar el schedule

    # Early Stopping
    'early_stopping': False,
    'patience': 50,                # (inactivo: early_stopping=False)

    # Datos
    'activations_file': Path(f"../../../activations/data/layer{NAMING_META['capa']}_{NAMING_I
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}
```

```python
# Crear directorio local del modelo
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['batch_size', 'num_epochs', 'learning_rate', 'warmup_steps', 'sparsity_warmup_ste
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['p_start', 'p_end', 'n_sparsity_updates', 'anneal_start', 'estimated_epochs']:
    config_data['Categoría'].append('P-Annealing')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)
```

```
=====================================================================
 CONFIGURACIÓN DEL SAE
=====================================================================
     Categoría             Parámetro Valor
  Arquitectura             input_dim   512
  Arquitectura            hidden_dim  8192
  Arquitectura          tied_weights False
 Entrenamiento            batch_size  2048
 Entrenamiento            num_epochs   200
 Entrenamiento         learning_rate 0.001
 Entrenamiento          warmup_steps  8000
 Entrenamiento sparsity_warmup_steps 16000
 Entrenamiento         sparsity_coef   0.5
   P-Annealing               p_start   1.0
   P-Annealing                 p_end   0.2
   P-Annealing    n_sparsity_updates    10
   P-Annealing          anneal_start 40000
   P-Annealing      estimated_epochs   200
Early Stopping        early_stopping False
Early Stopping              patience    50
=====================================================================

Archivo de activaciones: ..\..\..\activations\data\layer6_100000games.npy
Expansión: 16.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standa
annealing_standard\100000games\ex4\plots
=====================================================================
```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```python
class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
```

```
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})
print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')
print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)


# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
```

```python
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False,  # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)
```

```
==================================================
 DATASET DE ACTIVACIONES
==================================================
        Propiedad          Valor
   Total muestras      5,900,000
            Shape (5900000, 512)
            Dtype        float32
Tamaño en memoria    11523.44 MB
==================================================

==================================================
 SPLIT DE DATOS (80-20)
==================================================
  Conjunto  Muestras Porcentaje
     Train 4,720,000      80.0%
Validation 1,180,000      20.0%
     Total 5,900,000     100.0%
==================================================

==================================================
 DATALOADERS
==================================================
DataLoader  Batches  Batch Size Shuffle
     Train     2305        2048      Sí
Validation      577        2048      No
==================================================
```

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder**: Linear → ReLU (sparsity natural)- **Decoder**: Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)
```

```python
# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim']  # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '-'
    ],
    'Parámetros': [
        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)
```

```
======================================================================
 ARQUITECTURA DEL MODELO
```

```
======================================================================
     Componente  Dimensión Parámetros
          Input        512          -
        Encoder 512 → 8192  4,202,496
Hidden (Latent)       8192          -
        Decoder 8192 → 512  4,194,304
         Output        512          -
          Total          -  8,397,312
======================================================================

 Factor de expansión: 16.0x
 Tied weights: No
 Device: cuda
======================================================================
```

6. Función de Pérdida

Loss = MSE(reconstrucción) + × L1(activaciones)

- **MSE**: Mide qué tan bien reconstruimos las activaciones originales
- **L1**: Penaliza activaciones grandes → fuerza sparsity

```python
class ConstrainedAdam(torch.optim.Adam):
    """
    Adam donde las columnas de decoder se proyectan a norma unitaria en cada step.
    Antes del paso: elimina la componente del gradiente paralela al decoder.
    Después del paso: renormaliza las columnas del decoder a norma 1.
    """
    def __init__(self, params, constrained_params, lr, betas=(0.9, 0.999)):
        super().__init__(params, lr=lr, betas=betas)
        self.constrained_params = list(constrained_params)

    def step(self, closure=None):
        with torch.no_grad():
            for p in self.constrained_params:
                normed_p = p / p.norm(dim=0, keepdim=True)
                p.grad -= (p.grad * normed_p).sum(dim=0, keepdim=True) * normed_p
        super().step(closure=closure)
        with torch.no_grad():
            for p in self.constrained_params:
                p /= p.norm(dim=0, keepdim=True)
```

```python
def loss_function(x, reconstruction, hidden, sparsity_coeff, p, step=0, sparsity_warmup_steps
    """
    Pérdida compuesta: MSE (sum/mean) + Lp^p penalty con sparsity warmup

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse, >= 0 por ReLU)
        sparsity_coeff: Peso adaptativo de la penalización (alpha)
        p: Exponente actual del schedule de annealing
        step: Step global actual (para sparsity warmup)
        sparsity_warmup_steps: Steps de rampa de sparsity (0 = desactivado)
    """
    # BUG2 FIX: sum sobre features luego mean sobre batch (igual que referencia)
    mse_loss = (reconstruction - x).pow(2).sum(dim=-1).mean()
    # BUG4 FIX: sparsity warmup - rampa lineal de 0 a 1
    sparsity_scale = min(step / sparsity_warmup_steps, 1.0) if sparsity_warmup_steps > 0 else
    sparsity_loss = hidden.pow(p).sum(dim=1).mean()   # Lp^p: Σ|f_i|^p
    total_loss = mse_loss + sparsity_coeff * sparsity_scale * sparsity_loss
    return total_loss, mse_loss, sparsity_loss


# ConstrainedAdam: pasa model.decoder.parameters() como parámetros constrained
optimizer = ConstrainedAdam(
    model.parameters(),
    model.decoder.parameters(),
    lr=CONFIG['learning_rate']
)

# BUG3 FIX: LR warmup lineal - sube de 0 a lr en warmup_steps
def _lr_lambda(step):
    warmup = CONFIG.get('warmup_steps', 1000)
    return min(step / warmup, 1.0) if warmup > 0 else 1.0

scheduler = torch.optim.lr_scheduler.LambdaLR(optimizer, lr_lambda=_lr_lambda)

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'p_start', 'p_en
    'Valor': [
        'ConstrainedAdam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
```

```python
        f"{CONFIG['p_start']}",
        f"{CONFIG['p_end']}",
        f"{CONFIG['n_sparsity_updates']}",
        f"{CONFIG['anneal_start']}"
    ]
})

print('\n' + '='*55)
print(' OPTIMIZACIÓN')
print('='*55)
print(optim_info.to_string(index=False))
print('='*55)
```

```
=======================================================
 OPTIMIZACIÓN
=======================================================
              Parámetro           Valor
            Optimizador ConstrainedAdam
          Learning Rate           1e-03
Sparsity Coef (inicial)           5e-01
                p_start             1.0
                  p_end             0.2
     n_sparsity_updates              10
           anneal_start           40000
=======================================================
```

7. Loop de Entrenamiento

```python
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
```

```
    'val_fraction_active': [],
    'sparsity_coef_history': [],   # snapshot de sparsity_coeff al final de cada época
    'p_history': [],               # snapshot de p al final de cada época
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

# ========== P-ANNEALING STATE ==========
p = CONFIG['p_start']
next_p = None
p_step_count = 0
sparsity_coeff = CONFIG['sparsity_coef']
sparsity_queue = []
sparsity_queue_length = 10

total_steps = CONFIG['num_epochs'] * len(train_loader)
# BUG5 FIX: usar expected_steps (no total_steps) para que el schedule
# de annealing complete antes del early stopping
estimated_epochs = CONFIG.get('estimated_epochs', 50)
expected_steps = estimated_epochs * len(train_loader)
_update_steps_tensor = torch.linspace(
    CONFIG['anneal_start'], expected_steps, CONFIG['n_sparsity_updates'], dtype=torch.int
)
sparsity_update_steps_set = set(_update_steps_tensor.tolist())
sparsity_update_steps_list = sorted(sparsity_update_steps_set)
p_values = torch.linspace(CONFIG['p_start'], CONFIG['p_end'], CONFIG['n_sparsity_updates']).t
global_step = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' P-annealing: p {CONFIG["p_start"]} → {CONFIG["p_end"]} en {CONFIG["n_sparsity_update
print(f'   Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):

    # ========== ENTRENAMIENTO ==========
    model.train()

    epoch_total_loss = 0
```

```
    epoch_mse_loss = 0
    epoch_sparsity_loss = 0
    epoch_l0 = 0
    epoch_fraction_active = 0

    pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] p={p:.3f

    for batch in pbar:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # ---- P-ANNEALING: lookahead para la cola ----
        with torch.no_grad():
            if next_p is not None:
                lp_curr = hidden.pow(p).sum(dim=1).mean().item()
                lp_next = hidden.pow(next_p).sum(dim=1).mean().item()
                sparsity_queue.append([lp_curr, lp_next])
                sparsity_queue = sparsity_queue[-sparsity_queue_length:]

            # Verificar si este step es un punto de actualización de p
            if (global_step in sparsity_update_steps_set
                    and p_step_count < CONFIG['n_sparsity_updates']
                    and global_step >= sparsity_update_steps_list[p_step_count]):
                # Re-escalar sparsity_coeff para compensar el cambio de magnitud de Lp^p
                if next_p is not None and len(sparsity_queue) > 0:
                    local_curr = sum(i[0] for i in sparsity_queue) / len(sparsity_queue)
                    local_next = sum(i[1] for i in sparsity_queue) / len(sparsity_queue)
                    if local_next > 0:
                        sparsity_coeff = sparsity_coeff * (local_curr / local_next)
                # Actualizar p al siguiente valor del schedule
                p = p_values[p_step_count]
                if p_step_count < CONFIG['n_sparsity_updates'] - 1:
                    next_p = p_values[p_step_count + 1]
                else:
                    next_p = CONFIG['p_end']
                p_step_count += 1

        # Calcular pérdida con p actual
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, sparsity_coeff, p,
```

```python
            step=global_step,
            sparsity_warmup_steps=CONFIG.get('sparsity_warmup_steps', 0)
        )

        # Backward pass
        optimizer.zero_grad()
        total_loss.backward()
        optimizer.step()
        scheduler.step()  # BUG3 FIX: LR warmup step

        global_step += 1

        # Métricas
        with torch.no_grad():
            l0 = (hidden > 0).float().sum(dim=1).mean().item()
            fraction_active = (hidden > 0).float().mean().item()

        epoch_total_loss += total_loss.item()
        epoch_mse_loss += mse_loss.item()
        epoch_sparsity_loss += sparsity_loss.item()
        epoch_l0 += l0
        epoch_fraction_active += fraction_active

        pbar.set_postfix({
            'loss': f'{total_loss.item():.4f}',
            'mse': f'{mse_loss.item():.4f}',
            'l0': f'{l0:.1f}',
            'p': f'{p:.3f}'
        })

    # Promedios de entrenamiento
    num_train_batches = len(train_loader)
    avg_train_total_loss = epoch_total_loss / num_train_batches
    avg_train_mse_loss = epoch_mse_loss / num_train_batches
    avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
    avg_train_l0 = epoch_l0 / num_train_batches
    avg_train_fraction_active = epoch_fraction_active / num_train_batches

    # ========== VALIDACIÓN ==========
    model.eval()

    val_total_loss = 0
```

```python
    val_mse_loss = 0
    val_sparsity_loss = 0
    val_l0 = 0
    val_fraction_active = 0

    with torch.no_grad():
        for batch in val_loader:
            x = batch.to(device)
            reconstruction, hidden = model(x)
            total_loss, mse_loss, sparsity_loss = loss_function(
                x, reconstruction, hidden, sparsity_coeff, p,
                step=global_step,
                sparsity_warmup_steps=CONFIG.get('sparsity_warmup_steps', 0)
            )
            l0 = (hidden > 0).float().sum(dim=1).mean().item()
            fraction_active = (hidden > 0).float().mean().item()

            val_total_loss += total_loss.item()
            val_mse_loss += mse_loss.item()
            val_sparsity_loss += sparsity_loss.item()
            val_l0 += l0
            val_fraction_active += fraction_active

    num_val_batches = len(val_loader)
    avg_val_total_loss = val_total_loss / num_val_batches
    avg_val_mse_loss = val_mse_loss / num_val_batches
    avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
    avg_val_l0 = val_l0 / num_val_batches
    avg_val_fraction_active = val_fraction_active / num_val_batches

    # Guardar en historial
    history['train_total_loss'].append(avg_train_total_loss)
    history['train_mse_loss'].append(avg_train_mse_loss)
    history['train_sparsity_loss'].append(avg_train_sparsity_loss)
    history['train_l0_sparsity'].append(avg_train_l0)
    history['train_fraction_active'].append(avg_train_fraction_active)

    history['val_total_loss'].append(avg_val_total_loss)
    history['val_mse_loss'].append(avg_val_mse_loss)
    history['val_sparsity_loss'].append(avg_val_sparsity_loss)
    history['val_l0_sparsity'].append(avg_val_l0)
    history['val_fraction_active'].append(avg_val_fraction_active)
```

```python
history['sparsity_coef_history'].append(sparsity_coeff)
history['p_history'].append(p)

# ========== EARLY STOPPING ==========
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

        best_checkpoint = {
            'model_state_dict': model.state_dict(),
            'optimizer_state_dict': optimizer.state_dict(),
            'config': CONFIG,
            'history': history,
            'epoch': best_epoch,
            'val_mse': best_val_mse
        }

        best_model_name = get_model_name(
            variante=NAMING_META['variante'],
            tecnica=NAMING_META['tecnica'],
            capa=NAMING_META['capa'],
            juegos=NAMING_META['juegos'],
            estado='best'
        )
        torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

        save_checkpoint(
            model,
            base_path=NAMING_META['base_path'],
            variante=NAMING_META['variante'],
            tecnica=NAMING_META['tecnica'],
            capa=NAMING_META['capa'],
            juegos=NAMING_META['juegos'],
            estado='best',
            extras=NAMING_META['extras'],
            config=CONFIG,
            history=history,
            optimizer=optimizer
        )
```

```
            print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f
        else:
            epochs_no_improve += 1
            if epochs_no_improve >= CONFIG['patience']:
                print(f'\n  Early stopping activado en época {epoch+1}')
                print(f'   Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}
                break

    # Log cada 5 épocas
    if (epoch + 1) % 5 == 0 or epoch == 0:
        print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | p: {p:.3f} |  : {sparsity_coeff:.
        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0
        if CONFIG['early_stopping']:
            print(f'   Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})
```

```
print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)
```

```
 Iniciando entrenamiento: 200 épocas máximo
 P-annealing: p 1.0 → 0.2 en 10 saltos
  Early stopping: patience=50 épocas
======================================================================

Época 1/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [05:08<00:00,  7.47it/s, loss=2

 Época 1/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 17.0526 | MSE: 5.9338 | L0: 4463.3
  Val   - Total: 23.3960 | MSE: 5.5261 | L0: 3281.7

Época 2/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [04:29<00:00,  8.55it/s, loss=3
Época 3/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [06:23<00:00,  6.01it/s, loss=4
Época 4/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:23<00:00, 27.48it/s, loss=5
Época 5/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:21<00:00, 28.15it/s, loss=6

 Época 5/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 62.1809 | MSE: 18.9812 | L0: 908.5
  Val   - Total: 66.9169 | MSE: 21.2608 | L0: 836.1

Época 6/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 28.05it/s, loss=7
Época 7/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:21<00:00, 28.26it/s, loss=8
Época 8/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 28.07it/s, loss=8
Época 9/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 27.85it/s, loss=8
Época 10/200 [Train] p=1.000 =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 27.95it/s, loss=

 Época 10/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 82.3941 | MSE: 30.3209 | L0: 588.9
  Val   - Total: 82.5155 | MSE: 30.4406 | L0: 588.9
```

```
Época 11/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 28.08it/s, loss=
Época 12/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 27.97it/s, loss=
Época 13/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:21<00:00, 28.12it/s, loss=
Época 14/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 28.00it/s, loss=
Época 15/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 28.10it/s, loss=


 Época 15/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 82.0025 | MSE: 30.4294 | L0: 577.8
  Val   - Total: 82.2055 | MSE: 30.7347 | L0: 577.8

Época 16/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:21<00:00, 28.12it/s, loss=
Época 17/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:22<00:00, 28.00it/s, loss=
Época 18/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.29it/s, loss=
Época 19/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:25<00:00, 27.03it/s, loss=
Época 20/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.14it/s, loss=


 Época 20/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 81.8369 | MSE: 30.4604 | L0: 573.4
  Val   - Total: 82.0605 | MSE: 30.6213 | L0: 574.6

Época 21/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.21it/s, loss=
Época 22/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.17it/s, loss=
Época 23/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.25it/s, loss=
Época 24/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.12it/s, loss=
Época 25/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:25<00:00, 27.07it/s, loss=


 Época 25/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 81.7384 | MSE: 30.4740 | L0: 570.9
  Val   - Total: 81.9638 | MSE: 30.6311 | L0: 572.5

Época 26/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.17it/s, loss=
Época 27/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.23it/s, loss=
Época 28/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:25<00:00, 27.11it/s, loss=
Época 29/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:25<00:00, 27.09it/s, loss=
Época 30/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:25<00:00, 27.06it/s, loss=
```

```
 Época 30/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 81.6750 | MSE: 30.4788 | L0: 569.4
  Val   - Total: 81.9077 | MSE: 30.7607 | L0: 569.6

Época 31/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.19it/s, loss=
Época 32/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.21it/s, loss=
Época 33/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.26it/s, loss=
Época 34/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.23it/s, loss=
Época 35/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.27it/s, loss=


 Época 35/200 | p: 1.000 |  : 5.00e-01
  Train - Total: 81.6311 | MSE: 30.4916 | L0: 568.0
  Val   - Total: 81.9047 | MSE: 30.6440 | L0: 570.0

Época 36/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:26<00:00, 26.78it/s, loss=
Época 37/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:24<00:00, 27.22it/s, loss=
Época 38/200 [Train] p=1.000  =5.00e-01: 100%|     | 2305/2305 [01:25<00:00, 26.94it/s, loss=
Época 39/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.94it/s, loss=
Época 40/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.95it/s, loss=


 Época 40/200 | p: 0.911 |  : 4.50e-01
  Train - Total: 81.8251 | MSE: 33.1870 | L0: 466.4
  Val   - Total: 82.1194 | MSE: 33.6877 | L0: 464.1

Época 41/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:24<00:00, 27.19it/s, loss=
Época 42/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:24<00:00, 27.20it/s, loss=
Época 43/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.99it/s, loss=
Época 44/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.93it/s, loss=
Época 45/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.94it/s, loss=


 Época 45/200 | p: 0.911 |  : 4.50e-01
  Train - Total: 81.8094 | MSE: 33.2977 | L0: 460.4
  Val   - Total: 82.1287 | MSE: 33.6111 | L0: 461.2
```

```
Época 46/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 27.00it/s, loss=
Época 47/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.92it/s, loss=
Época 48/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.92it/s, loss=
Época 49/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.97it/s, loss=
Época 50/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 27.01it/s, loss=


 Época 50/200 | p: 0.911 |  : 4.50e-01
  Train - Total: 81.7776 | MSE: 33.2903 | L0: 459.1
  Val   - Total: 82.0855 | MSE: 33.6522 | L0: 460.0

Época 51/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 27.07it/s, loss=
Época 52/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.93it/s, loss=
Época 53/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.90it/s, loss=
Época 54/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:26<00:00, 26.79it/s, loss=
Época 55/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.83it/s, loss=


 Época 55/200 | p: 0.911 |  : 4.50e-01
  Train - Total: 81.7503 | MSE: 33.2788 | L0: 458.4
  Val   - Total: 82.0568 | MSE: 33.6101 | L0: 459.3

Época 56/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:26<00:00, 26.74it/s, loss=
Época 57/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.95it/s, loss=
Época 58/200 [Train] p=0.911  =4.50e-01: 100%|     | 2305/2305 [01:25<00:00, 26.83it/s, loss=
Época 59/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.90it/s, loss=
Época 60/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 27.00it/s, loss=


 Época 60/200 | p: 0.822 |  : 4.08e-01
  Train - Total: 81.9215 | MSE: 35.6571 | L0: 388.1
  Val   - Total: 82.2810 | MSE: 35.9827 | L0: 387.0

Época 61/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.97it/s, loss=
Época 62/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.85it/s, loss=
Época 63/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 27.05it/s, loss=
Época 64/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.89it/s, loss=
Época 65/200 [Train] p=0.822  =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.56it/s, loss=
```

```
 Época 65/200 | p: 0.822 |  : 4.08e-01
  Train - Total: 81.9308 | MSE: 35.7741 | L0: 380.0
  Val   - Total: 82.3271 | MSE: 36.1088 | L0: 381.1

Época 66/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.94it/s, loss=
Época 67/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.61it/s, loss=
Época 68/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.65it/s, loss=
Época 69/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.86it/s, loss=
Época 70/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.67it/s, loss=


 Época 70/200 | p: 0.822 |  : 4.08e-01
  Train - Total: 81.8881 | MSE: 35.6699 | L0: 378.9
  Val   - Total: 82.2510 | MSE: 36.1420 | L0: 378.9

Época 71/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.80it/s, loss=
Época 72/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.91it/s, loss=
Época 73/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.64it/s, loss=
Época 74/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.61it/s, loss=
Época 75/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 26.91it/s, loss=


 Época 75/200 | p: 0.822 |  : 4.08e-01
  Train - Total: 81.8557 | MSE: 35.5898 | L0: 378.2
  Val   - Total: 82.2497 | MSE: 35.9210 | L0: 379.4

Época 76/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:25<00:00, 27.05it/s, loss=
Época 77/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.68it/s, loss=
Época 78/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.79it/s, loss=
Época 79/200 [Train] p=0.822 =4.08e-01: 100%|     | 2305/2305 [01:26<00:00, 26.73it/s, loss=
Época 80/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.91it/s, loss=


 Época 80/200 | p: 0.733 |  : 3.73e-01
  Train - Total: 81.9452 | MSE: 37.3072 | L0: 329.0
  Val   - Total: 82.3472 | MSE: 37.6810 | L0: 326.7
```

```
Época 81/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.87it/s, loss=
Época 82/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.82it/s, loss=
Época 83/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.98it/s, loss=
Época 84/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.93it/s, loss=
Época 85/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:26<00:00, 26.76it/s, loss=


 Época 85/200 | p: 0.733 |  : 3.73e-01
  Train - Total: 81.8539 | MSE: 36.7851 | L0: 318.3
  Val   - Total: 82.2424 | MSE: 37.1256 | L0: 318.4

Época 86/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 27.02it/s, loss=
Época 87/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:24<00:00, 27.14it/s, loss=
Época 88/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 27.03it/s, loss=
Época 89/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:24<00:00, 27.22it/s, loss=
Época 90/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:24<00:00, 27.18it/s, loss=


 Época 90/200 | p: 0.733 |  : 3.73e-01
  Train - Total: 81.7260 | MSE: 36.0586 | L0: 315.7
  Val   - Total: 82.1164 | MSE: 36.3509 | L0: 316.4

Época 91/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:26<00:00, 26.79it/s, loss=
Época 92/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 27.05it/s, loss=
Época 93/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 27.07it/s, loss=
Época 94/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 27.05it/s, loss=
Época 95/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.81it/s, loss=


 Época 95/200 | p: 0.733 |  : 3.73e-01
  Train - Total: 81.6086 | MSE: 35.4808 | L0: 314.1
  Val   - Total: 82.0057 | MSE: 35.8138 | L0: 314.7

Época 96/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 26.97it/s, loss=
Época 97/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:26<00:00, 26.72it/s, loss=
Época 98/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:25<00:00, 27.03it/s, loss=
Época 99/200 [Train] p=0.733 =3.73e-01: 100%|     | 2305/2305 [01:24<00:00, 27.25it/s, loss=
Época 100/200 [Train] p=0.644 =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.85it/s, loss
```

```
 Época 100/200 | p: 0.644 |  : 3.47e-01
  Train - Total: 81.6739 | MSE: 36.3537 | L0: 280.1
  Val   - Total: 82.0640 | MSE: 36.6959 | L0: 277.7

Época 101/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.92it/s, loss
Época 102/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.82it/s, loss
Época 103/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:26<00:00, 26.77it/s, loss
Época 104/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.82it/s, loss
Época 105/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 27.08it/s, loss


 Época 105/200 | p: 0.644 |  : 3.47e-01
  Train - Total: 81.3104 | MSE: 34.9617 | L0: 271.4
  Val   - Total: 81.7098 | MSE: 35.1793 | L0: 272.0

Época 106/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 27.03it/s, loss
Época 107/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 27.01it/s, loss
Época 108/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.92it/s, loss
Época 109/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:26<00:00, 26.66it/s, loss
Época 110/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 27.01it/s, loss


 Época 110/200 | p: 0.644 |  : 3.47e-01
  Train - Total: 81.0813 | MSE: 34.1598 | L0: 269.7
  Val   - Total: 81.4967 | MSE: 34.6215 | L0: 269.5

Época 111/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:26<00:00, 26.80it/s, loss
Época 112/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:24<00:00, 27.19it/s, loss
Época 113/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.84it/s, loss
Época 114/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.82it/s, loss
Época 115/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:24<00:00, 27.22it/s, loss


 Época 115/200 | p: 0.644 |  : 3.47e-01
  Train - Total: 80.9304 | MSE: 33.7000 | L0: 269.2
  Val   - Total: 81.3907 | MSE: 34.2153 | L0: 269.0
```

```
Época 116/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.87it/s, loss
Época 117/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.93it/s, loss
Época 118/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:26<00:00, 26.73it/s, loss
Época 119/200 [Train] p=0.644  =3.47e-01: 100%|     | 2305/2305 [01:25<00:00, 26.85it/s, loss
Época 120/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.92it/s, loss


 Época 120/200 | p: 0.556 |  : 3.27e-01
  Train - Total: 81.4524 | MSE: 35.4931 | L0: 247.5
  Val   - Total: 81.8457 | MSE: 35.9774 | L0: 245.1

Época 121/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.82it/s, loss
Época 122/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:26<00:00, 26.75it/s, loss
Época 123/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:26<00:00, 26.79it/s, loss
Época 124/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:24<00:00, 27.37it/s, loss
Época 125/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:24<00:00, 27.38it/s, loss


 Época 125/200 | p: 0.556 |  : 3.27e-01
  Train - Total: 81.3017 | MSE: 35.5819 | L0: 240.0
  Val   - Total: 81.7946 | MSE: 36.0768 | L0: 240.1

Época 126/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:24<00:00, 27.39it/s, loss
Época 127/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:24<00:00, 27.31it/s, loss
Época 128/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.81it/s, loss
Época 129/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:24<00:00, 27.37it/s, loss
Época 130/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 27.04it/s, loss


 Época 130/200 | p: 0.556 |  : 3.27e-01
  Train - Total: 81.1954 | MSE: 35.3212 | L0: 238.9
  Val   - Total: 81.6856 | MSE: 35.8202 | L0: 239.2

Época 131/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.86it/s, loss
Época 132/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.99it/s, loss
Época 133/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.97it/s, loss
Época 134/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:28<00:00, 26.09it/s, loss
Época 135/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 27.04it/s, loss
```

```
 Época 135/200 | p: 0.556 |  : 3.27e-01
  Train - Total: 81.0991 | MSE: 35.0829 | L0: 238.5
  Val   - Total: 81.5953 | MSE: 35.6031 | L0: 238.7

Época 136/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 27.01it/s, loss
Época 137/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.90it/s, loss
Época 138/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.85it/s, loss
Época 139/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 27.00it/s, loss
Época 140/200 [Train] p=0.556  =3.27e-01: 100%|     | 2305/2305 [01:25<00:00, 26.94it/s, loss


 Época 140/200 | p: 0.467 |  : 3.09e-01
  Train - Total: 82.1122 | MSE: 37.2677 | L0: 225.3
  Val   - Total: 82.6831 | MSE: 38.4624 | L0: 219.4

Época 141/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.87it/s, loss
Época 142/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.82it/s, loss
Época 143/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.97it/s, loss
Época 144/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:26<00:00, 26.52it/s, loss
Época 145/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:24<00:00, 27.25it/s, loss


 Época 145/200 | p: 0.467 |  : 3.09e-01
  Train - Total: 82.0018 | MSE: 37.6574 | L0: 216.8
  Val   - Total: 82.5672 | MSE: 38.1960 | L0: 216.9

Época 146/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:29<00:00, 25.80it/s, loss
Época 147/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.80it/s, loss
Época 148/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:27<00:00, 26.34it/s, loss
Época 149/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:24<00:00, 27.17it/s, loss
Época 150/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 27.07it/s, loss


 Época 150/200 | p: 0.467 |  : 3.09e-01
  Train - Total: 81.8867 | MSE: 37.3572 | L0: 216.5
  Val   - Total: 82.3695 | MSE: 37.9259 | L0: 216.2
```

```
Época 151/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:24<00:00, 27.23it/s, loss
Época 152/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 27.01it/s, loss
Época 153/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 27.00it/s, loss
Época 154/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 27.10it/s, loss
Época 155/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:26<00:00, 26.64it/s, loss


 Época 155/200 | p: 0.467 |  : 3.09e-01
  Train - Total: 81.7533 | MSE: 36.9638 | L0: 216.8
  Val   - Total: 82.2793 | MSE: 37.5101 | L0: 216.8

Época 156/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:24<00:00, 27.14it/s, loss
Época 157/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:24<00:00, 27.28it/s, loss
Época 158/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.88it/s, loss
Época 159/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.96it/s, loss
Época 160/200 [Train] p=0.467  =3.09e-01: 100%|     | 2305/2305 [01:25<00:00, 26.92it/s, loss


 Época 160/200 | p: 0.378 |  : 2.93e-01
  Train - Total: 82.7571 | MSE: 38.6097 | L0: 210.6
  Val   - Total: 83.9338 | MSE: 40.6493 | L0: 203.5

Época 161/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 27.00it/s, loss
Época 162/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.95it/s, loss
Época 163/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.81it/s, loss
Época 164/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:26<00:00, 26.78it/s, loss
Época 165/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.85it/s, loss


 Época 165/200 | p: 0.378 |  : 2.93e-01
  Train - Total: 83.3321 | MSE: 40.1760 | L0: 200.1
  Val   - Total: 83.7614 | MSE: 40.6530 | L0: 199.8

Época 166/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:26<00:00, 26.70it/s, loss
Época 167/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.86it/s, loss
Época 168/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:24<00:00, 27.19it/s, loss
Época 169/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:26<00:00, 26.74it/s, loss
Época 170/200 [Train] p=0.378  =2.93e-01: 100%|     | 2305/2305 [01:29<00:00, 25.83it/s, loss
```

```
 Época 170/200 | p: 0.378 |  : 2.93e-01
  Train - Total: 83.2551 | MSE: 39.9831 | L0: 199.9
  Val   - Total: 83.8587 | MSE: 40.6482 | L0: 199.6

Época 171/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:36<00:00, 23.86it/s, loss
Época 172/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:24<00:00, 27.32it/s, loss
Época 173/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.91it/s, loss
Época 174/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:26<00:00, 26.59it/s, loss
Época 175/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:26<00:00, 26.71it/s, loss


 Época 175/200 | p: 0.378 |  : 2.93e-01
  Train - Total: 83.1240 | MSE: 39.6160 | L0: 200.6
  Val   - Total: 83.6551 | MSE: 40.2244 | L0: 200.2

Época 176/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:27<00:00, 26.29it/s, loss
Época 177/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.87it/s, loss
Época 178/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.90it/s, loss
Época 179/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:25<00:00, 26.95it/s, loss
Época 180/200 [Train] p=0.378 =2.93e-01: 100%|     | 2305/2305 [01:26<00:00, 26.74it/s, loss


 Época 180/200 | p: 0.289 |  : 2.78e-01
  Train - Total: 83.5741 | MSE: 40.0024 | L0: 199.8
  Val   - Total: 86.0044 | MSE: 43.5823 | L0: 192.2

Época 181/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 26.99it/s, loss
Época 182/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 26.81it/s, loss
Época 183/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 26.89it/s, loss
Época 184/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:27<00:00, 26.39it/s, loss
Época 185/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:26<00:00, 26.64it/s, loss


 Época 185/200 | p: 0.289 |  : 2.78e-01
  Train - Total: 85.6510 | MSE: 43.7917 | L0: 187.3
  Val   - Total: 86.1092 | MSE: 44.4216 | L0: 186.4
```

```
Época 186/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:26<00:00, 26.65it/s, loss
Época 187/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.30it/s, loss
Época 188/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 27.11it/s, loss
Época 189/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 26.96it/s, loss
Época 190/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 27.06it/s, loss


 Época 190/200 | p: 0.289 |  : 2.78e-01
  Train - Total: 85.5212 | MSE: 43.5577 | L0: 187.3
  Val   - Total: 86.0246 | MSE: 44.0803 | L0: 187.1

Época 191/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.18it/s, loss
Época 192/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.26it/s, loss
Época 193/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.24it/s, loss
Época 194/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.15it/s, loss
Época 195/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:25<00:00, 26.99it/s, loss


 Época 195/200 | p: 0.289 |  : 2.78e-01
  Train - Total: 85.4996 | MSE: 43.4636 | L0: 187.3
  Val   - Total: 85.8302 | MSE: 43.8444 | L0: 187.1

Época 196/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.23it/s, loss
Época 197/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:26<00:00, 26.55it/s, loss
Época 198/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:30<00:00, 25.47it/s, loss
Época 199/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.13it/s, loss
Época 200/200 [Train] p=0.289 =2.78e-01: 100%|     | 2305/2305 [01:24<00:00, 27.13it/s, loss


 Época 200/200 | p: 0.289 |  : 2.78e-01
  Train - Total: 85.4668 | MSE: 43.3544 | L0: 187.4
  Val   - Total: 85.9509 | MSE: 43.8666 | L0: 187.3

======================================================================
 RESUMEN DEL ENTRENAMIENTO
======================================================================
Entrenamiento completo: 200 épocas

Métricas Finales (Última Época):
        Métrica   Train Validation
     Total Loss 85.4668    85.9509
```

```
       MSE Loss 43.3544    43.8666
    L0 Sparsity   187.4      187.3
Fraction Active   2.29%      2.29%
======================================================================
```

```
import torch
ck = torch.load('saved_model/sae_standard_p-annealing_l6_12500g_final.pt', weights_only=Fals
print(ck['history']['p_history'])
```

```
[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1
```

8. Guardar Modelo

```
# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
```

```
    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)
```

```
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
annealing_standard\100000games\sae_standard_p-annealing_l6_100000g_ex4_final.pt

======================================================================
 CHECKPOINTS GUARDADOS
======================================================================
      Tipo                                               Ubicación
     Local saved_model\sae_standard_p-annealing_l6_100000g_final.pt
Organizado                                     layer_06/models/...
======================================================================
```

9. Métricas básicas del modelo

```
# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
```

```
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "p_start": CONFIG["p_start"],
    "p_end": CONFIG["p_end"],
    "n_sparsity_updates": CONFIG["n_sparsity_updates"],
    "anneal_start": CONFIG["anneal_start"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print("" + "="*50)
```

```
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)
```

```
==================================================
 METRICAS DE ENTRENAMIENTO GUARDADAS
==================================================
        Metrica     Valor
Val MSE (final) 43.866551
Val MSE (mejor)       inf
 Val L0 (final)     187.3
Fraccion Activa     2.29%
Archivo: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\
annealing_standard\100000games\ex4\metrics\training_metrics.json
==================================================
```

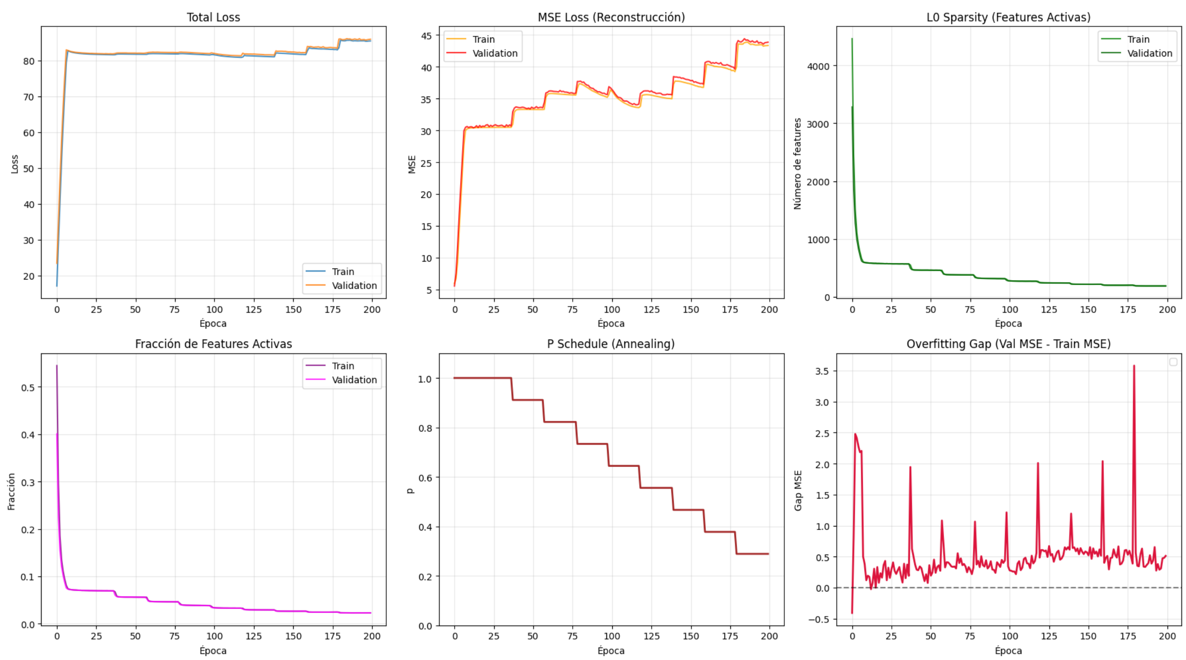
10. Visualización de Resultados

```
# Gráficas de pérdidas (Train vs Validation) + P-annealing schedule
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Bes
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
```

```
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# P schedule (annealing)
axes[1, 1].plot(history['p_history'], color='brown', linewidth=2, label='p')
axes[1, 1].set_title('P Schedule (Annealing)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('p')
axes[1, 1].set_ylim(0, 1.1)
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Bes
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
```

```
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('   Gráficas guardadas exitosamente')
print('='*50)
```

```
C:\Users\Esposa\AppData\Local\Temp\ipykernel_14784\2801488273.py:59: UserWarning: No artists
  axes[1, 2].legend()
```




```
==================================================
   Gráficas guardadas exitosamente
==================================================
```

11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
```

```python
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:,}"
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos
```

```python
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)
print('   ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)
```

```
======================================================================
 RESUMEN FINAL DEL ENTRENAMIENTO
======================================================================

Arquitectura:
Componente        Valor
     Input          512
    Hidden 8192 (16.0x)
Parámetros    8,397,312

Datos:
  Conjunto  Muestras
     Train 4,720,000
Validation 1,180,000
     Total 5,900,000

Métricas Finales:
   Métrica   Valor
Train Loss 85.4668
  Val Loss 85.9509
 Train MSE 43.3544
   Val MSE 43.8666
  Train L0   187.4
    Val L0   187.3

Archivos Guardados:
    Tipo
```

```
  Modelo
annealing_l6_100000g_final.pt
Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\
annealing_standard\100000games\ex4\plots\training_curves.png

======================================================================
   ENTRENAMIENTO COMPLETADO EXITOSAMENTE
======================================================================
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_pannealing.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
```

```python
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sa
annealing_standard\100000games\ex4\sae_standard_p-annealing_l6_100000g_ex4_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\lay
annealing_standard\100000games\ex4\sae_standard_p-annealing_l6_100000g_ex4_training.pdf
```